

Ultra-stable lasers and photonic integrated circuits for secure communications

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Outline

1. Motivation

2. Theory

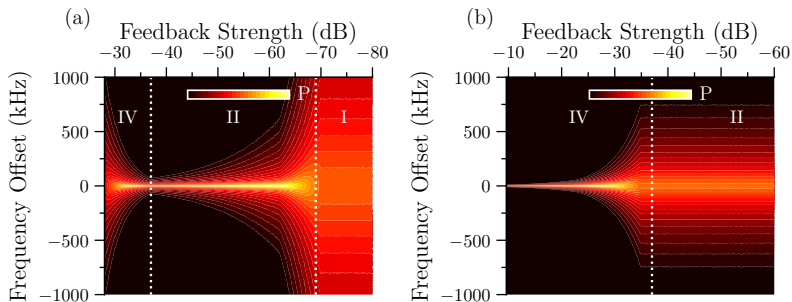
3. Method

4. Results

- Free-running characteristics
- Characteristics under external optical feedback

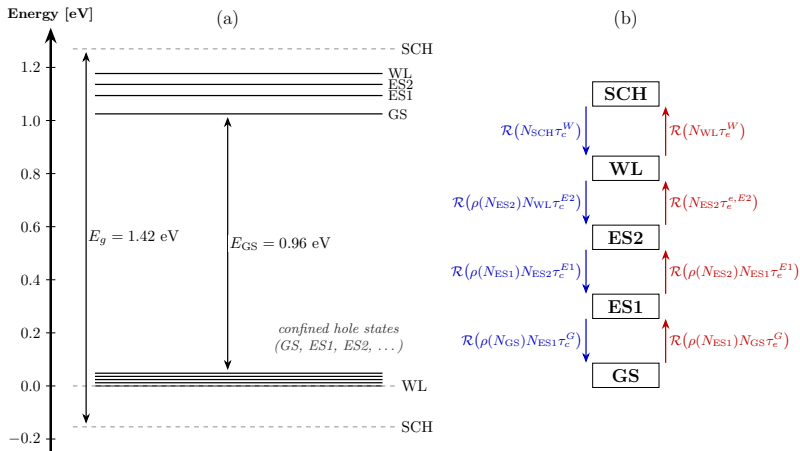
5. Outlook & Conclusion

Motivation



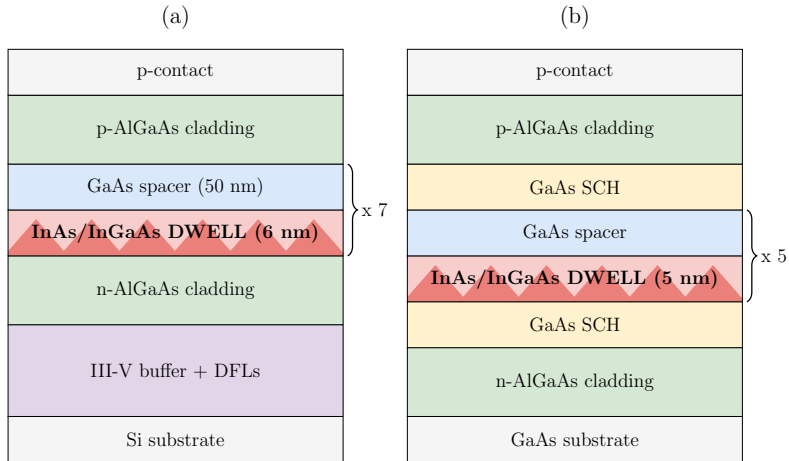
Simulated power spectral density as a function of feedback strength for a (a) QW laser and (b) QD laser. The data used in this plot is taken from [1].

Theory - QD Models



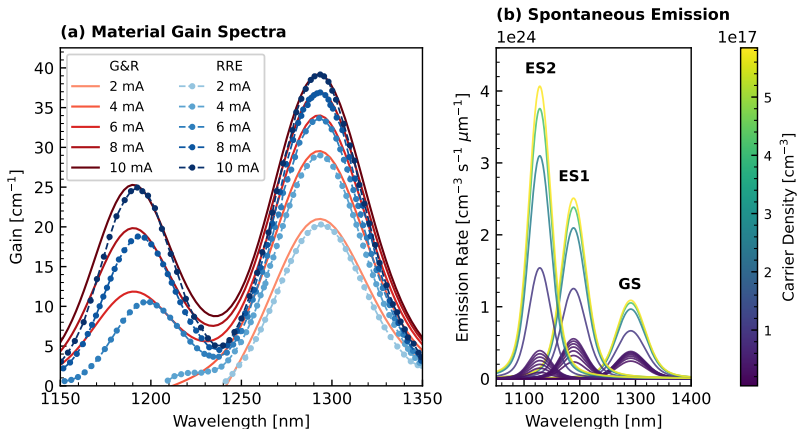
Cascade model described by Gioannini and Rosseti (2011) [2].

QD DFB Structure



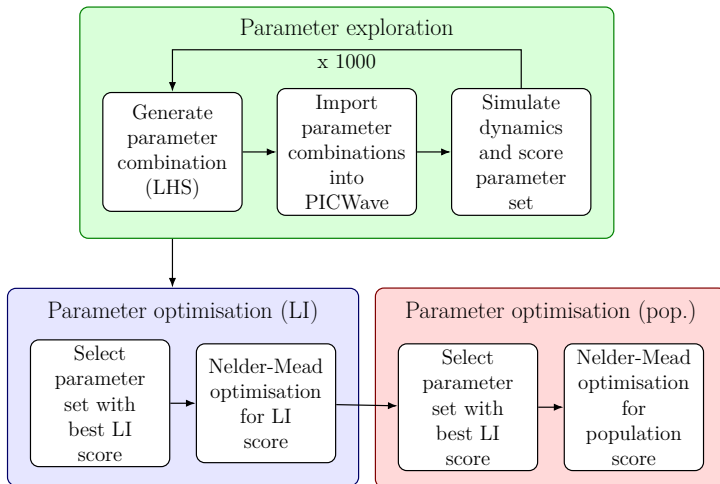
(a) QD DFB structure grown by UCL researchers (Wang et al. 2008) [3] (b) QD DFB structure simulated by (Gioannini et al 2011) [2], experimentally implemented by [4].

Optical properties



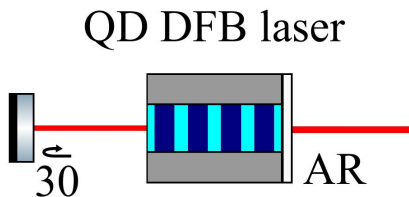
(a) Comparison between the gain curves obtained by G&R [2] and this work (b) Spontaneous emission spectrum in this work. calculated in this work using similar parameters

Method - Parameter optimisation



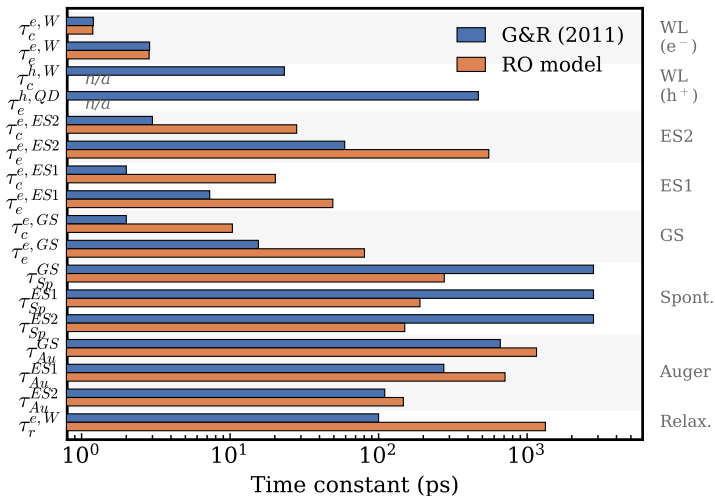
Flowchart showing the three stages used to find an optimal parameter set.

Laser model



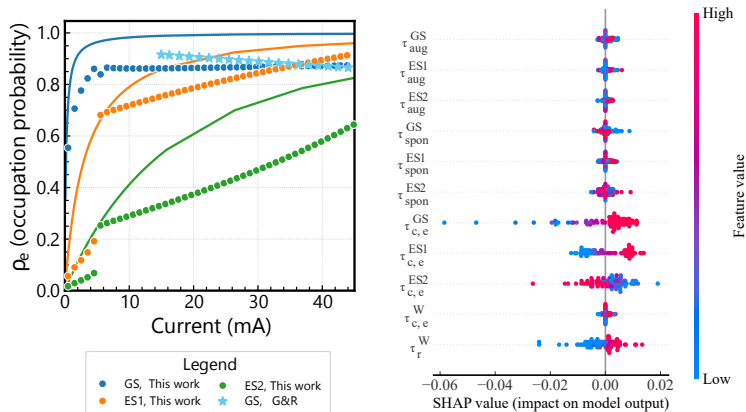
Model of the QD DFB circuit simulated in PICWave.

Results - Parameter optimisation



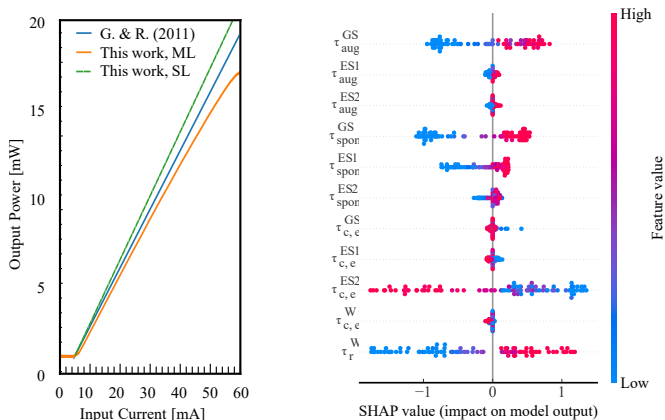
Comparison between the parameter set in the G&R model [2], and the optimised parameter set found in this work.

Population dynamics



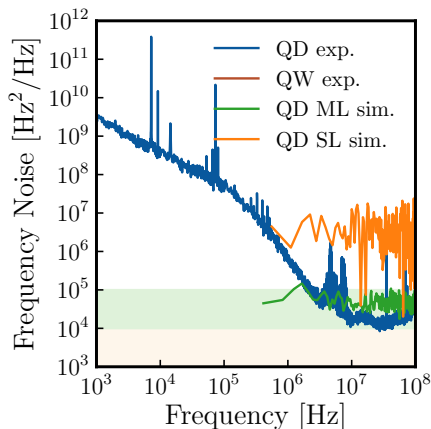
(a) Occupation probability for quantum dot states as a function of input current. (b) "Beeswarm" plot, which presents the SHAP values across parameters. This shows the relative influence of each parameter on the occupation score

Power-Current characteristics



(a) Output power against input current comparing G&R [2], and the multilevel (ML) and single level (SL) models from this work. (b) "Beeswarm" plot, which presents the SHAP values across parameters.

Frequency noise & linewidth

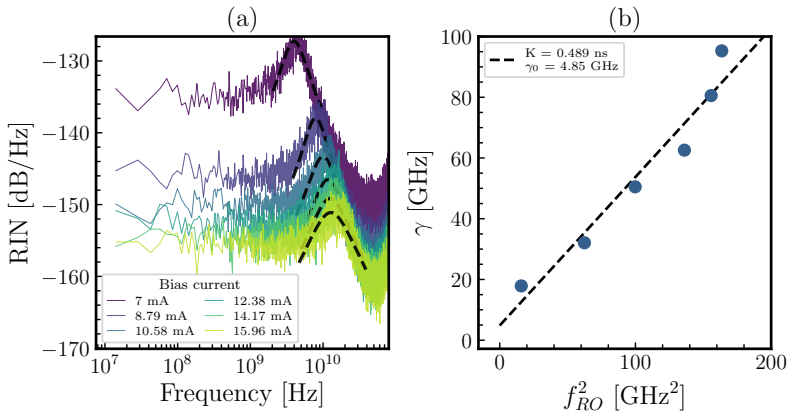


Frequency noise spectrum comparing data from [1] and this work.

Study	Linewidth
This work (ml.)	160 kHz
This work (sl.)	16 MHz
Dong et al. [1]	61 kHz
Wang et al. [5]	243 kHz
Wan et al. [6]	157 kHz

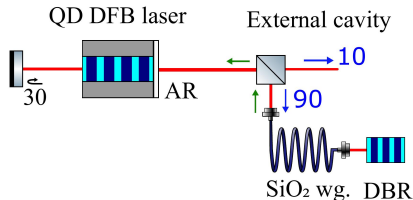
Comparison of free-running QD laser linewidths.

RIN & Damping factor

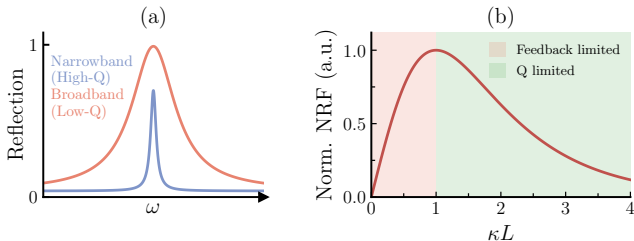


(a) Plot of RIN spectrum with a Lorentzian peak (b) Plot of γ^2 against f_{RO}^2 . The K value given by the slope of the best fit line was 0.489 ns, and the γ_0 value, the y intercept, was 4.85 GHz.

Behaviour under EOF (1)

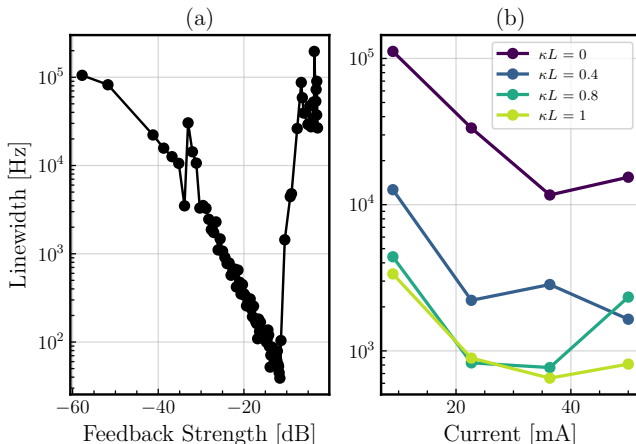


PICWave simulation schematic of the QD DFB laser under external optical feedback.



(a) Narrowband vs. broadband reflector response. (b) Normalized noise reduction factor as a function of κL .

Behaviour under EOF (2)



(a) Linewidth plotted against feedback strength, which is controlled by the grating strength. (b) Linewidth against current current as a function of grating strength.

Summary: Strengths & Limitations

Positives	Negatives
Accurate linewidth - within experimental range.	Unphysical parameters differ from experimental values.
Optical feedback resilience to -11 dB demonstrated.	Overestimated resonance frequency.
GP surrogate model shows effect of parameters on characteristics.	Gain compression overestimated.
Flexibility of PICWave demonstrated with EOF demonstration.	Thermal effects/flicker noise not currently implemented.

Future work

- ▶ More sophisticated cost function - e.g. rewards other physical effects
- ▶ Incorporate temperature modulation - possible in PICWave
- ▶ Longer waveguide in EOF experiment
- ▶ Adjust quality of QD distribution - e.g. inhomogeneous linewidth broadening

References I



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